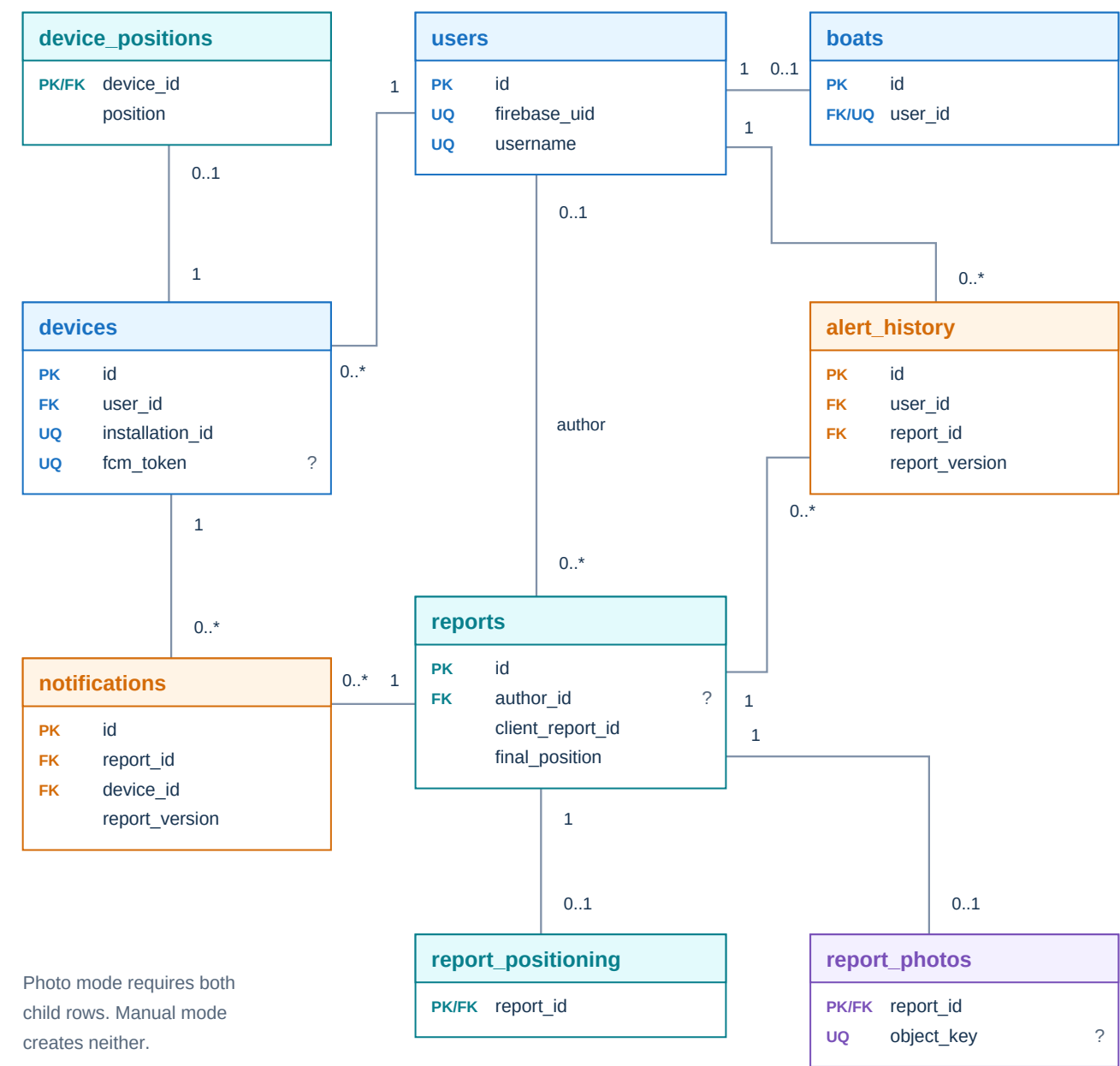


BlueWay database schema

Relational key view | 10 tables | 14 foreign keys

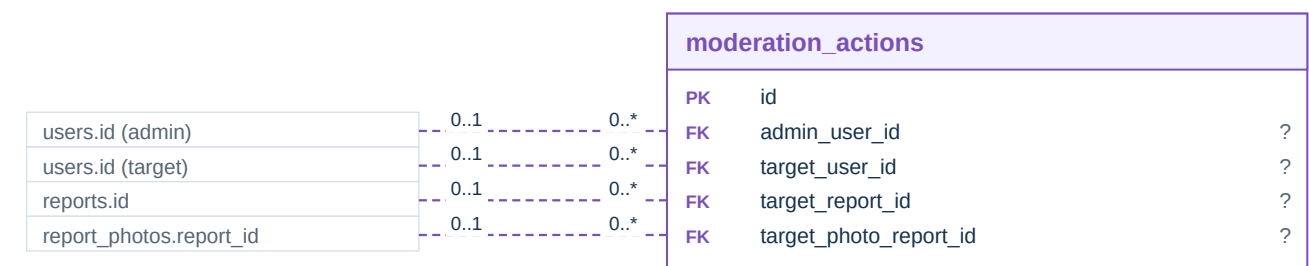
1 = one 0..1 = optional one 0..* = zero or more ? = nullable

PK = primary key; FK = foreign key; UQ = unique. Core deletes: CASCADE, except author_id -> SET NULL.



Moderation and audit

References below point to the same tables above. Audit foreign keys use SET NULL.



Complete fields and SQL types follow. Validation and deletion rules are on pages 4-5.

Database fields - identity and reports

View A: 47 fields across five tables. Field types and nullability are shown.

users		
Field	Type	Key / null
id	uuid	PK
firebase_uid	text	UQ
username	varchar(100)	UQ
email	varchar(254)	UQ ?
date_of_birth	date	?
nationality	char(2)	?
role	varchar(10)	-
status	varchar(10)	-
show_user_name	boolean	-
show_boat_info	boolean	-
notifications_enabled	boolean	-
created_at	timestamp	-
updated_at	timestamp	-
No password or authentication token is stored.		

boats		
Field	Type	Key / null
id	uuid	PK
user_id	uuid	FK/UQ
name	varchar(100)	?
boat_type	varchar(20)	-
flag_country	char(2)	?
created_at	timestamp	-
updated_at	timestamp	-
user_id -> users.id		

device_positions		
Field	Type	Key / null
device_id	uuid	PK/FK
position	GeoPoint	-
accuracy_m	double	-
heading_deg	double	?
measured_at	timestamp	-
received_at	timestamp	-
device_id -> devices.id		

reports		
Field	Type	Key / null
id	uuid	PK
author_id	uuid	FK ?
client_report_id	uuid	-
category	varchar(20)	-
positioning_mode	varchar(10)	-
description	varchar(250)	?
final_position	GeoPoint	-
observed_at	timestamp	-
created_at	timestamp	-
expires_at	timestamp	-
status	varchar(10)	-
version	integer	-
updated_at	timestamp	-
removed_at	timestamp	?
author_id -> users.id (nullable after deletion)		
UQ: author_id + client_report_id		

devices		
Field	Type	Key / null
id	uuid	PK
user_id	uuid	FK
installation_id	uuid	UQ
fc_m_token	text	UQ ?
platform	varchar(10)	-
is_active	boolean	-
last_seen_at	timestamp	-
user_id -> users.id		
installation_id and fcm_token are unique		
Owner stays fixed; account switch -> new UUID.		

Reports are separate from device positions. final_position is the map point and the origin for recipient selection.

One latest usable position per device. No position-age cutoff. Validation: pages 4-5.

Database fields - photos alerts and audit

View B: 52 fields across five tables. Field types and nullability are shown.

report_positioning		
Field	Type	Key / null
report_id	uuid	PK/FK
observer_position	GeoPoint	-
gps_accuracy_m	double	-
azimuth_deg	double	-
inclination_deg	double	-
camera_height_m	numeric(6,2)	-
camera_height_source	varchar(30)	?
camera_height_uncertainty_m	double	?
focal_length_mm	double	?
zoom_ratio	double	?
estimated_position	GeoPoint	?
estimated_distance_m	double	?
algorithm_version	varchar(50)	?
captured_at	timestamp	-
report_id -> reports.id		

report_photos		
Field	Type	Key / null
report_id	uuid	PK/FK
upload_status	varchar(10)	-
object_key	text	UQ ?
size_bytes	integer	?
mime_type	varchar(100)	?
uploaded_at	timestamp	?
hidden_at	timestamp	?
created_at	timestamp	-
updated_at	timestamp	-
report_id -> reports.id		

GeoPoint = geography(Point,4326). Coordinate order: longitude, latitude. timestamp = timestamptz; double = double precision.

Photo mode needs both left-hand tables; manual mode has neither. State-dependent validation: pages 4-5.

notifications		
Field	Type	Key / null
id	uuid	PK
report_id	uuid	FK
device_id	uuid	FK
report_version	integer	-
status	varchar(10)	-
attempt_count	integer	-
next_attempt_at	timestamp	?
sent_at	timestamp	?
last_error_code	text	?
created_at	timestamp	-
updated_at	timestamp	-
report_id -> reports.id; device_id -> devices.id UQ: report_id + device_id + report_version		

alert_history		
Field	Type	Key / null
id	uuid	PK
user_id	uuid	FK
report_id	uuid	FK
report_version	integer	-
category	varchar(20)	-
position	GeoPoint	-
distance_m	double	-
alerted_at	timestamp	-
user_id -> users.id; report_id -> reports.id UQ: user_id + report_id + report_version		

moderation_actions		
Field	Type	Key / null
id	uuid	PK
admin_user_id	uuid	FK ?
target_user_id	uuid	FK ?
target_report_id	uuid	FK ?
target_photo_report_id	uuid	FK ?
action	varchar(20)	-
reason	text	-
previous_status	varchar(10)	?
new_status	varchar(10)	?
created_at	timestamp	-

Audit FKs may become NULL after deletion. New actions need an active admin and one target.

Database integrity rules

These rules complete the 99-field dictionary on pages 2-3. Reference:

Technical-Documentation-escali.docx, database dictionary and additional integrity rules. The ten application tables and fourteen foreign keys are unchanged.

Keys and deletion

The primary key of each table is required. Unique keys: users.firebase_uid, users.username, users.email when present; boats.user_id; devices.installation_id and devices.fcm_token when present; report_photos.object_key when present.

Composite uniqueness: reports(author_id, client_report_id); notifications(report_id, device_id, report_version); alert_history(user_id, report_id, report_version). report_version is a historical number, not a foreign key to reports.version.

Boat, device, position, photo, positioning, notification and recipient-history rows follow parent deletion through CASCADE. reports.author_id and the four moderation_actions foreign keys use SET NULL. An author is required when a report is created. Account deletion preserves the report.

Required values and defaults

users.role: user | admin. users.status: active | suspended. devices.platform: android | ios. reports.positioning_mode: manual | photo. reports.status: active | removed | expired.

Report and history category: marine_animal | obstruction | pollution. Photo upload status: pending | uploaded | failed. Notification status: pending | sent | failed. Boat type: voilier | bateau_moteur | catamaran | semi_rigide | jet_ski | autre.

show_user_name, show_boat_info and notifications_enabled default to false. reports.version defaults to 1; notifications.attempt_count defaults to 0. Both report_version fields and reports.version are at least 1; attempt_count is non-negative.

Measurements and spatial indexes

All measurements must be finite when present. Both GPS accuracy fields are between 0 and 50 metres inclusive. heading_deg and azimuth_deg are at least 0 and less than 360. Inclination is 0 at the horizon, positive upwards and negative downwards; the reference imposes no further angular range.

Camera height is strictly positive. Optional height uncertainty and estimated distance are non-negative; optional focal length and zoom are strictly positive. alert_history.distance_m is non-negative.

All geographic fields are geography(Point,4326), longitude before latitude. All event times are timestampz. GiST indexes cover reports.final_position and device_positions.position. Additional indexes cover reports.author_id, reports(status, expires_at), devices.user_id and notifications(status, next_attempt_at).

Transactions and lifecycle

Photo consistency and upload

A photo report has exactly one `report_positioning` row and one `report_photos` row. A manual report has neither. Save the report and its children in one transaction; deferred constraint triggers check their final state at commit, including child deletion and reassignment.

Photo bytes remain in Scaleway Object Storage. The database stores metadata only. A pending or failed upload does not invalidate the report. Retry using the same `report_id` and photo row. Uploaded means that the JPEG and its storage metadata have been saved. A hidden photo keeps its upload state.

When present, `size_bytes` is greater than 0 and at most 500000; `mime_type` is `image/jpeg`. The backend must verify actual JPEG content, successful decoding and absence of EXIF. SQL metadata constraints cannot validate the object bytes.

Reports and devices

`expires_at` equals `observed_at` plus 24 hours. Editing never extends the original lifetime. Restoration clears `removed_at`. Map visibility requires `status = active` and `expires_at` later than current server time; an expired report need not be rewritten by a scheduled job.

The application rejects future observations and observations older than 24 hours at publication. A published edit increments version. Authentication, ownership, allowed edits and active-account checks belong to the backend service.

Keep only the latest usable device position. The device owner remains fixed. On account change, clear the old token, position and queued tasks in one transaction, coordinate with the notification worker, and register a new installation UUID.

Alerts and audit

Select other active users who opted into proximity alerts, with an active device, an FCM token and a usable position within 18520 metres. Heading and position age do not exclude recipients. Store the minimum eligible-device distance in each user history entry.

Save history and per-device notification tasks in the report transaction. Pending notifications require `next_attempt_at`; sent notifications require `sent_at`. The worker processes committed tasks and rechecks eligibility, version and expiry. FCM acceptance does not prove delivery; history survives push failure.

Moderation actions: `remove_report` and `restore_report` target a report; `hide_photo` targets `report_photos`; `suspend_user` and `reactivate_user` target a user. At insertion, require an active administrator, exactly one matching target and a non-empty reason. Save the action and audit together. Audit content cannot be edited; parent deletion may clear its links.

Only these ten business tables are persistent application entities. Map tiles and display clusters are generated, not stored as additional tables. Database constraints support transaction integrity; recipient selection, upload handling and moderation workflows remain backend responsibilities.